

Reviewer Pack — Minimal Number of Binary Operations in Automata Bounds ACC^0 ...

Entry 68b4925b35b3 · INCONCLUSIVE

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Minimal Number of Binary Operations in Automata Bounds ACC^0 Circuit Size

Entry ID: 68b4925b35b3 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-28 12:19:37 UTC

1. Conjecture

Field A (mathematical branch): Automata Theory **Field B** (complexity object): Complexity Theory: ACC^0 Circuit Complexity

Statement:

[‘For every explicit function $f: \{0, \dots, N-1\} \rightarrow R$ in P , the minimal number of binary operations required by any deterministic finite automaton (DFA) recognizing the language L_f generated by f is $\Theta(N \log N)$.’, ‘Equivalently, there exists a polynomial-time algorithm that, given an explicit function f in P and its corresponding DFA A with minimal binary operations, can determine whether A recognizes L_f or not.’]

Rationale (proposer’s reasoning):

[‘This conjecture bridges Automata Theory with ACC^0 Circuit Complexity by proposing a natural property of deterministic automata that could serve as a potential barrier to AC^0 lower bounds.’, ‘Automata theory provides a simple model of computation, while the minimal number of binary operations in an automaton is a fundamental invariant that could potentially relate to circuit complexity.’, ‘If true, this conjecture would suggest that certain properties inherent to deterministic automata are closely linked to the complexities of explicit functions.’]

Taxonomy category: ACC_SIPSER (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 1495930a57a6c637

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all 30 independent random seeds, the number of binary operations in the minimal DFA is within $\Theta(N \log N)$, and the polynomial-time algorithm correctly determines recognition for all generated functions.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	1.00	UNCERTAIN	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "deterministic finite automaton" AND "ACC⁰ circuit complexity" - "minimal number of binary operations" AND automata - "language recognition by DFA" AND "polynomial-time algorithm"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_function(n):
        return [random.randint(0, 1) for _ in range(n)]

    def construct_dfa(f):
        n = len(f)
        states = list(range(2 * n + 1))
        transitions = {}
        accepting_states = set()

        for i in range(n):
            if f[i] == 0:
                accepting_states.add(i + n)

        for q in states:
            for a in [0, 1]:
                if q < n and a == f[q]:
                    transitions[(q, a)] = q + 1
                elif q >= n and (a == 0 or q - n == i):
                    transitions[(q, a)] = q
                else:
                    transitions[(q, a)] = q

        return states, transitions, accepting_states

    def count_binary_operations(transitions):
```

```

        return sum(len(v) for v in transitions.values())

n = random.randint(5, 40)
f = generate_function(n)
states, transitions, _ = construct_dfa(f)
binary_operations = count_binary_operations(transitions)

metric_name = 'binary_operations'
metric_value = binary_operations
instances_tested = 1
conjecture_holds = binary_operations == n * (n + 1) // 2
counterexample = "mapping_undefined" if not conjecture_holds else ""

return {
    "metric_name": metric_name,
    "metric_value": metric_value,
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i - 1 for i in range(5, 30)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = (sum((r["metric_value"] - mean_value)**2 for r in results) / len(results))**0.5
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        print(f"RESULT: FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_bfcc5240.py", line 73, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_bfcc5240.py", line 51, in run_trial

```

```

    binary_operations = count_binary_operations(transitions)
                        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_bfcc5240.py", line 46, in count_binary_
    return sum(len(v) for v in transitions.values())
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_bfcc5240.py", line 46, in <genexpr>
    return sum(len(v) for v in transitions.values())
                ^^^^^^
TypeError: object of type 'int' has no len()

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing data, which means that the pre-registered support conditions could not be verified. | next: Re-run the test to ensure it completes successfully and verify if the pre-registered support conditions are met.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11666
2	preregistration	ollama_remote	glm4:latest	0	0	5486
3	novelty	ollama_remote	glm4:latest	0	0	4669
4	novelty	ollama_remote	glm4:latest	0	0	7742
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	41142
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8611
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8201
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8209
9	judge	ollama_remote	glm4:latest	0	0	13437

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 109161 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/68b4925b35b3.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/68b4925b35b3.jsonl` for verification of provenance

4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/68b4925b35b3.tar.gz` (if generated)